LUMINOSITIES

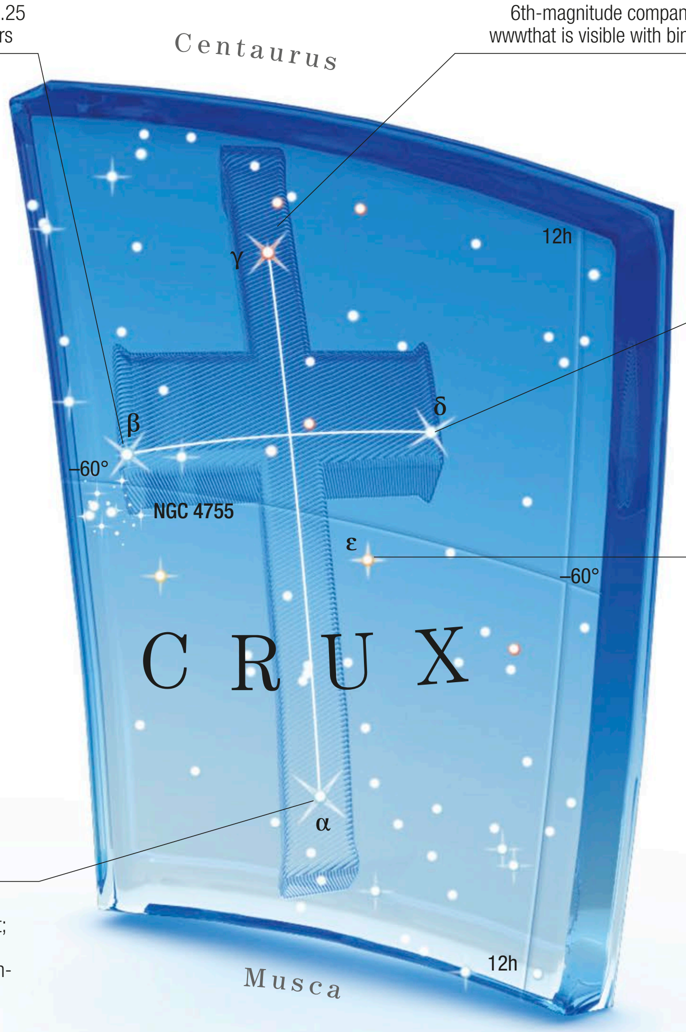
Gamma Crucis *148 Suns*

Epsilon Crucis *158 Suns*

CRUX
THE SOUTHERN CROSS

ALTHOUGH IT IS THE SMALLEST CONSTELLATION OF ALL, CRUX IS ONE OF THE MOST DISTINCTIVE DUE TO ITS FOUR BRIGHT STARS. CROSSED BY THE MILKY WAY'S STAR-RICH PATH, IT HOSTS ONE OF THE GEMS OF THE SOUTHERN NIGHT SKY: THE JEWEL BOX CLUSTER.

Situated between the legs of Centaurus, Crux is the sky's most compact grouping of four bright stars. Its brilliant stars were known to the ancient Greeks but only mapped as a separate constellation in the 16th century. Crux first appeared in its modern form on the celestial globe of cartographer Petrus Plancius in 1598. Initially called Crux Australis, the Southern Cross, it is now known simply as Crux. The southern end of its cross-shaped pattern is marked by the constellation's brightest star, Acrux. It, Mimosa, and Gacrux are in the top 25 brightest night-time stars. More distant than Crux's four main stars, at about 600 light-years away, is a wedge-shaped dark patch of sky named the Coalsack. This dark nebula of gas and dust is visible to the naked eye because it blocks out light from the dense Milky Way star fields behind it. Just north and about ten times more distant than the Coalsack is the Jewel Box Cluster (NGC 4755). This appears as a fuzzy star to the naked eye but binoculars reveal individual stars.



Mimosa (β **Crucis)**
A blue-white giant; also a variable that changes in magnitude between 1.25 and 1.35 every 6 hours

Gacrux (γ **Crucis)**
A red giant at least 85 times the width of the Sun; it has an unrelated 6th-magnitude companion star wwwthat is visible with binoculars

Delta (δ **Crucis)**
A blue-white star that is moving from the main-sequence to the red-giant stage of its life

Epsilon (ε) **Crucis**
An orange giant with about 1.4 times the Sun's mass and 33 times its width; it lies 230 light-years away and has a magnitude of 3.6

Acrux (α **Crucis)**
A blue-white subgiant; a telescope reveals it has a blue-white main-sequence companion star of magnitude 1.8

▷ **Locating the South Celestial Pole**
Crux has been used for centuries as a pointer to the South Celestial Pole. Its bright stars and the two brightest in Centaurus (Hadar and Alpha Centauri) are easy to spot, as can be seen in the photograph above. Extend southward a line connecting Gacrux and Acrux, and an imaginary line bisecting Alpha Centauri and Hadar. The two cross just east of the South Pole, the nearest star to which is Sigma Octantis in Octans.

KEY DATA

Size ranking 88
Brightest stars Acrux (α) 0.8, Mimosa (β) 1.25–1.35
Genitive Crucis
Abbreviation Cru
Highest in sky at 10pm April–May
Fully visible 25°N–90°S

CHART 2

MAIN STARS

Acrux Alpha (α) Crucis
Blue-white subgiant; also a double star
0.8 ⟷ 322 light-years

Mimosa Beta (β) Crucis
Blue-white giant; also a variable star
1.25–1.35 ⟷ 278 light-years

Gacrux Gamma (γ) Crucis
Red giant
1.6 ⟷ 89 light-years

Delta (δ) Crucis
Blue-white subgiant
2.8 ⟷ 345 light-years

DEEP-SKY OBJECTS

NGC 4755 (Jewel Box Cluster)
Open cluster

Coalsack Nebula
Dark nebula